ADVANCED HCI EXAMINATION

Human-Computer Interaction (CCS3204/CIT3441)

ZCAS University | School of Computing, Technology and Applied Sciences

Date: May 22, 2026

EXAMINATION INSTRUCTIONS

Total Marks: 100 marks

Duration: 3 hours

Answer ALL questions in Section A and ANY 3 questions from Section B.

SECTION A: SCENARIO-BASED BABY QUESTIONS (40 MARKS)

SCENARIO: A local Zambian e-commerce mobile application called 'ShopZam' allows users to browse and purchase products online. Recently, the app received complaints about usability issues. Users report confusion when navigating the app, difficulty finding products, unclear error messages when payments fail, and frustration with the slow loading times. The app was designed by a team without extensive user testing, and many design decisions were made without considering user feedback or user behavior patterns.

1. Define Human-Computer Interaction (HCI) in your own words.

2. List two main differences between usability and user experience.

3. Using the ShopZam scenario, identify one usability problem mentioned.

4. What is feedback in HCI? Give an example from ShopZam.

5. Define 'mental model' and explain how it relates to ShopZam users.

6. What is a 'system model'? How does it differ from a user's mental model?

7. Using Fitts' Law, explain why small buttons on mobile apps cause frustration.

8. List three HCI principles that should guide interface design.

9. What is cognitive load? How does it affect mobile app design?

10. Explain why slow loading times cause poor user experience in ShopZam.

11. Define 'consistency' in HCI and give a ShopZam example.

12. What is user frustration? List two causes visible in ShopZam.

13. Explain the importance of clear error messages in e-commerce apps.

14. What does 'user-centered design' mean? How would it help ShopZam?

15. List four elements of good user interface (UI) design.

16. What is accessibility in HCI? Name one accessibility concern for ShopZam.

17. Explain how cultural factors might affect ShopZam's design in Zambia.

18. Define 'discoverability'. Why is it important in ShopZam?

19. What is a 'use case' in HCI? Give one ShopZam use case.

20. Explain the relationship between usability and conversion rates in e-commerce.

21. What is 'intuitive design'? How should ShopZam improve intuitiveness?

22. List three ways poor design leads to user abandonment of apps.

23. Define 'affordance' in interface design with a ShopZam button example.

24. What is the difference between a wireframe and a prototype?

25. Explain 'visual hierarchy' and its importance in ShopZam's product pages.

26. What is 'information architecture'? How does it affect ShopZam navigation?

27. Define 'user testing' and explain why ShopZam needed it before launch.

28. What are 'heuristics' in HCI? Name two Nielsen's usability heuristics.

29. Explain the concept of 'progressive disclosure' in mobile interfaces.

30. What is 'response time' and how does it affect ShopZam user satisfaction?

31. List three mobile HCI considerations different from desktop design.

32. What is a 'user persona'? How would creating personas help ShopZam?

33. Explain 'learnability' as a usability metric for ShopZam.

34. What is the difference between task completion rate and user satisfaction?

35. Define 'interaction design' and give one ShopZam interaction example.

36. What is 'color theory' in UI design? Why is it important for ShopZam?

37. Explain the concept of 'dark patterns' and why they harm user trust.

38. What is 'responsive design'? Why should ShopZam implement it?

39. List three metrics used to measure usability in mobile applications.

40. Explain how user feedback and iteration improve the ShopZam app design.

SECTION B: STRUCTURED QUESTIONS - CHOOSE ANY 3 OF 4 (60 MARKS)

Each question carries 20 marks. Answer only 3 questions.

QUESTION 1 - HCI PRINCIPLES & USABILITY (20 MARKS)

SCENARIO: ShopZam's users report navigation confusion, unclear error messages, slow loading, and difficulty locating buttons. The design was created without user testing.

a) Identify and explain three HCI usability problems in ShopZam. (7 marks)

b) Using Nielsen's Usability Heuristics, discuss how ShopZam violates at least two heuristics and suggest improvements. (8 marks)

c) Explain how the concept of 'mental models' applies to ShopZam users and why alignment between mental model and system model is critical. (5 marks)

QUESTION 2 - USER-CENTERED DESIGN PROCESS (20 MARKS)

SCENARIO: ShopZam was designed without user input. The company now wants to redesign using a User-Centered Design (UCD) approach to address complaints.

a) Define the User-Centered Design (UCD) process and list four main stages. (7 marks)

b) Explain how user research and usability testing could have prevented ShopZam's problems before launch. (8 marks)

c) Describe two methods for gathering user feedback (e.g., interviews, surveys, user testing) and explain which would be most appropriate for ShopZam. (5 marks)

QUESTION 3 - MOBILE HCI vs DESKTOP HCI (20 MARKS)

SCENARIO: ShopZam exists as a mobile app. The company is considering building a desktop version but wants to understand design differences.

a) Compare mobile HCI and desktop HCI by discussing differences in screen size, input methods, and user behavior. (7 marks)

b) Using Fitts' Law and cognitive load concepts, explain why small buttons and dense menus harm mobile usability more than desktop usability. (8 marks)

c) Design three specific interface adjustments ShopZam should make when transitioning from mobile to desktop (e.g., button sizing, navigation structure, layout). (5 marks)

QUESTION 4 - ACCESSIBILITY, DIVERSITY & CULTURAL CONSIDERATIONS (20 MARKS)

SCENARIO: ShopZam wants to expand to serve elderly users, visually impaired users, and users across different Zambian regions with varying internet speeds and digital literacy.

a) Discuss three accessibility challenges for ShopZam and propose design solutions for visually impaired users. (7 marks)

b) Explain how cultural factors and socioeconomic differences in Zambia should influence ShopZam's design (e.g., language, payment methods, internet speed). (8 marks)

c) Propose two design features that would make ShopZam more usable for elderly users with limited digital experience. (5 marks)